Library Management System · UML Class Diagram

Visibility: + public · # protected · - private · underlined = class-level (static / classmethod)

models

«abstract» BaseEntity
- + id: UUID
- + *to_dict(): dict*

Role
- + SEED: tuple
- + name: str
- + description: str

User
- + ROLE_STUDENT: str
- + ROLE_LIBRARIAN: str
- + ROLE_ADMIN: str
- + ROLES: tuple
- + STAFF_ROLES: tuple
- + username: str
- + email: str
- + phone: str
- + role: str
- + is_active: bool
- + google_id: str
- + avatar_url: str
- + created_at: datetime
- − _password_hash: str

- + «set» password(plain: str)
- + check_password(plain: str): bool
- + is_staff_role(role: str): bool
- + is_admin_role(role: str): bool
- + is_admin(): bool
- + to_dict(): UserDict

Borrow
- + STATUS_PENDING_BORROW: str
- + STATUS_READY_FOR_PICKUP: str
- + STATUS_BORROWED: str
- + STATUS_PENDING_RETURN: str
- + STATUS_RETURNED: str
- + STATUS_REJECTED: str
- + STATUS_OVERDUE: str
- + STATUSES: tuple
- + OVERDUE_ELIGIBLE_STATUSES: tuple
- + RETURNABLE_STATUSES: tuple
- + user_id: UUID
- + book_id: UUID
- + status: str
- + borrowed_at: datetime
- + due_date: datetime
- + approved_at: datetime
- + returned_at: datetime
- + rejection_reason: str
- + created_at: datetime

- + is_overdue(): bool
- + approve(book: Book): void
- + confirm_pickup(loan_days: int): void
- + reject(reason: str): void
- + mark_returned(book: Book): void
- − _require_status(allowed: tuple, message: str): void
- − _isoformat_utc(value: datetime): str
- + to_dict(): BorrowDict

Book
- + isbn: str
- + title: str
- + author: str
- + category_id: UUID
- + description: str
- + pages: int
- + image_url: str
- + is_deleted: bool
- + archived_at: datetime
- + created_at: datetime
- − _total_copies: int
- − _available_copies: int

- + «get» total_copies: int
- + «get» available_copies: int
- + «get» average_rating: float
- + «get» total_reviews: int
- + adjust_total_copies(new_total: int): void
- + decrement_available(): void
- + increment_available(): void
- + to_dict(): BookDict

Category
- − _name: str

- + «get/set» name: str
- + to_dict(): CategoryDict

Review
- + MIN_RATING: int = 1
- + MAX_RATING: int = 5
- + book_id: UUID
- + user_id: UUID
- + rating: int
- + comment: str
- + created_at: datetime

- + parse_rating(raw_rating): int
- + to_dict(): ReviewDict

BorrowStatus
- + SEED: tuple
- + code: str
- + description: str

Favorite
- + user_id: UUID
- + book_id: UUID
- + created_at: datetime

- + to_dict(): FavoriteDict

Relationships
- Role 1 — 0..* User : *has role*
- User 1 — 0..* Borrow : *places*
- Borrow 0..* — 1 Book : *for*
- Book 0..* — 0..1 Category : *classifies*
- User 1 — 0..* Review : *writes*
- Book 1 — 0..* Review : *saves*
- Book 1 ◆— 1 Review : *has reviews*
- Borrow 0..* — 1 BorrowStatus : *in status*
- User 1 — 0..* Favorite : *saved in*
- Book 1 — 0..* Favorite : *saved in*

«use»

services

BaseService
- # _parse_uuid(val, param_name: str): UUID
- # _get_or_404(model, record_id, label: str): BaseEntity

BookService
- + DEFAULT_PAGE_SIZE: int = 10
- + MAX_PAGE_SIZE: int = 100

- + get_paginated_books(search_query, category_id, page, limit)
- + get_staff_catalog(state)
- + get_book_by_id(book_id, include_deleted)
- + create_book(form_data, image_file)
- + update_book(book_id, form_data, image_file)
- + archive_book(book_id)
- + restore_book(book_id)
- + delete_book(book_id)
- + get_related_books(book_id, limit)
- + get_or_create_category(category_input)
- + get_book_reviews(book_id)
- + get_book_reviews_stats(book_id)
- + add_review(book_id, user_id, rating, comment)
- + get_user_favorites(user_id)
- + add_favorite(user_id, book_id)
- + remove_favorite(user_id, book_id)
- − _parse_book_integer(raw_value, label, minimum)

BorrowService
- + BORROW_DURATION: int = 14
- + ACTIVE_STATUSES: tuple

- + get_all_borrows_admin()
- + get_borrowed_history(user_id)
- + get_user_active_borrows(user_id)
- + request_borrow(user_id, book_id)
- + approve_borrow_request(borrow_id)
- + confirm_borrow_pickup(borrow_id)
- + reject_borrow_request(borrow_id, reason)
- + approve_return_request(borrow_id)
- − _apply(borrow, change)

CategoryService
- + get_all_categories()
- + create_category(name)
- + update_category(category_id, new_name)
- + delete_category(category_id)

exceptions

«Python built-in» Exception

LibraryError
- + status_code: int = 500
- + message: str

- + __init__(message: str)
- + to_dict(): dict

«Python built-in» ValueError

ValidationError
- + status_code: int = 400

ConflictError
- + status_code: int = 409

NotFoundError
- + status_code: int = 404

«Python built-in» LookupError

RegistrationValidationError

RegistrationConflictError

DuplicateReviewError